

SMARTRECO HACKATHON · PRODUCT & ARCHITECTURE DOCUMENTATION

# Growise

## Behavioural AI recommendations for an online learning marketplace

A product walkthrough, implementation architecture, and reviewer guide

|                  |                    |                         |
|------------------|--------------------|-------------------------|
| EXPERIENCE       | RECOMMENDATIONS    | OPERATIONS              |
| Learner journeys | Grounded, personal | Auditable admin console |

*Prepared from the live local application experience and the current source implementation.*

01 · INTRODUCTION

# What Growise is

Growise is a course marketplace that replaces generic “related courses” with a behavioural AI recommendation loop. It observes the meaningful choices a learner makes—searches, course views, card clicks, qualified dwell time and enrollment intent—and converts those signals into a concise, catalog-grounded next-step recommendation.

The product is designed as two complementary experiences. Learners receive a focused route through a large course catalog, while administrators can manage catalog quality and inspect the evidence, agent steps, model usage and outcomes behind every recommendation.

# The problem

Traditional course discovery expects learners to articulate their goal, filter a large catalog and repeatedly evaluate alternatives. That creates three recurring problems: choice overload, recommendations that are not connected to current intent, and opaque systems that give operators no credible way to explain why a course was selected.

# What we are solving

Growise turns passive browsing into a lightweight learning signal. It makes a recommendation only when evidence exists, retrieves candidate courses from the real catalog, applies relevance and diversity checks, generates a tailored explanation through Mesh, and stores the result with an inspectable trace. The result is a more useful learner experience without treating every click as a reason to call an LLM.

| Design goal        | How Growise addresses it                                                                                                    |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Personal relevance | Weighted interest signals, recency and exclusions shape a learner profile before retrieval.                                 |
| Catalog grounding  | Chroma retrieves real course candidates; generated product IDs are validated against the candidate set.                     |
| Cost discipline    | A trigger gate serves the cached recommendation until there is enough new meaningful activity and the cooldown is complete. |
| Operational trust  | The admin console exposes events, recommendation traces, Mesh calls and vector-sync health.                                 |

02 · ARCHITECTURE

# Complete architecture

The frontend and backend are intentionally separated. Next.js provides server-rendered pages and interactive client components. FastAPI owns authentication, catalog mutations, behavioural event intake, recommendation delivery, operations data and the agent workflow. JWT bearer tokens keep cross-origin development simple; FastAPI enforces user and administrator boundaries on protected endpoints.

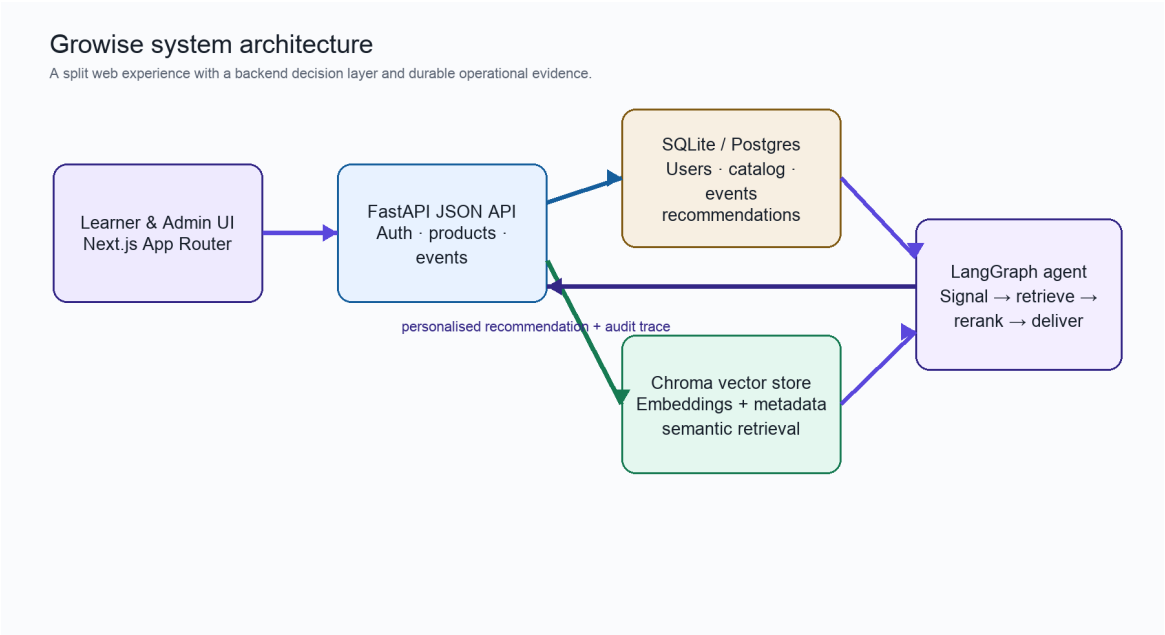


Figure 1. End-to-end Growise architecture. **Reviewer action:** Read left to right: the web UI records and requests; FastAPI coordinates durable data and the agent; the vector store supplies grounded candidates.

Component responsibilities

| Layer               | Responsibility                                                                               | Key implementation choices                                                      |
|---------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Next.js 16 frontend | Learner and admin journeys, course discovery, auth UI and event capture.                     | App Router, Tailwind UI, non-blocking tracker, API client.                      |
| FastAPI backend     | JSON API, JWT validation, administrative guards and orchestration.                           | Routers for auth, products, events, enrollments, recommendations and agent ops. |
| SQL store           | System of record for users, courses, events, enrollments, recommendations and run telemetry. | SQLAlchemy entities and durable recommendation history.                         |
| Chroma              | Semantic catalog retrieval.                                                                  | Course metadata + local all-MiniLM-L6-v2 embeddings; explicit sync status.      |
| LangGraph + Mesh    | Structured recommendation reasoning and persuasive narrative generation.                     | Bounded workflow, relevance checks, candidate validation and measured usage.    |

Behavioural intelligence and agent workflow

The tracker queues events in memory and flushes them every five seconds or at twenty events. It uses keepalive on page hide/unload so a final batch can complete without blocking navigation. Short visits remain available to operations, but only dwell at or above the qualifying threshold affects recommendation reasoning.

## Recommendation decision loop

The agent reasons from consented behavioural evidence and is grounded in the live catalog.

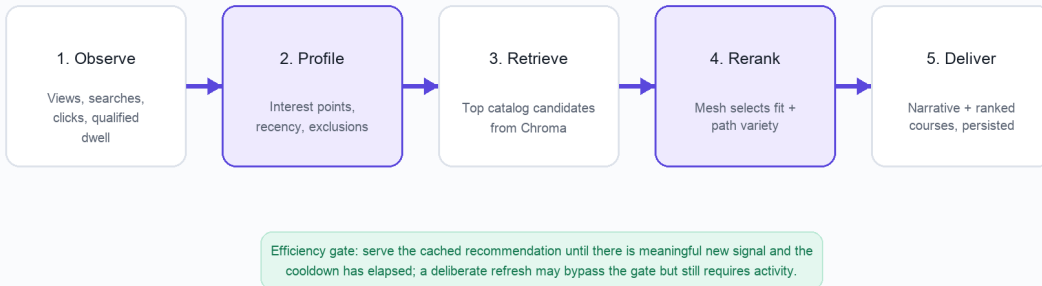


Figure 2. Behavioural signal to delivered recommendation. **Reviewer action:** Use this flow to verify the important boundary: an LLM never invents from an unconstrained catalog; it ranks and explains retrieved candidates.

In the profile stage, adjacent card-click and product-view events are collapsed into a single course-opening session. Interest is weighted by event type, dwell, enrollment intent and recency. Enrolled courses and recently dismissed recommendations become exclusions. Retrieval obtains a small candidate set from Chroma, relevance can be checked and refined once, and Mesh then reranks for fit and learning-path variety before the final narrative and ranked items are persisted.

## Catalog integrity and persistence

Course administration uses a deliberate dual-write pattern. A product is committed to the primary database before Chroma is updated. If semantic upsert fails, the course remains visible with `vector_synced=false` and the failure is retained for the catalog-health view. This avoids losing a valid catalog edit while making retrieval readiness explicit and safely retryable.

### Catalog dual-write and recommendation persistence

The catalog remains visible and auditable even if semantic sync needs a retry.

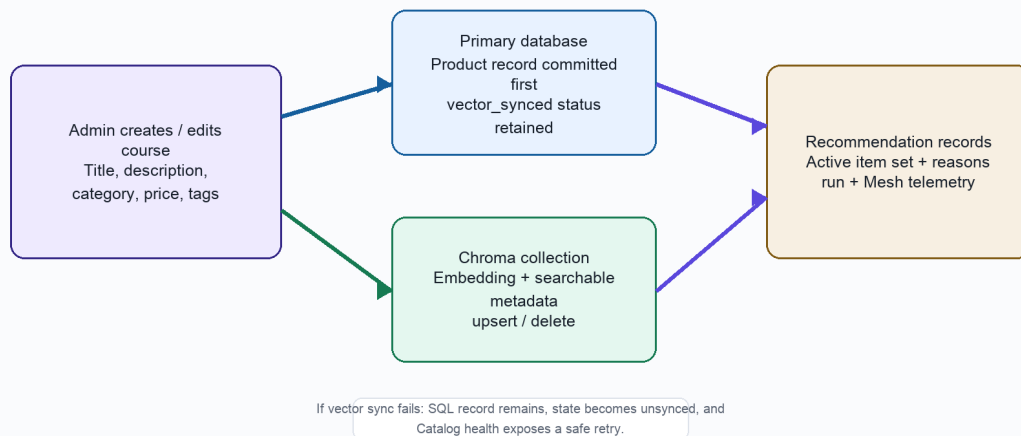


Figure 3. Dual-write catalog flow and recommendation evidence. **Reviewer action:** Verify that the primary record is committed first, then inspect Catalog health to see whether the semantic copy is ready or needs a retry.

### 03 · DATA, SECURITY AND EFFICIENCY

## Data model and lifecycle

| Record                         | Purpose                                                  | Lifecycle / guardrail                                                     |
|--------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------|
| users                          | Email/password credential, role and tracking preference. | Password hash; JWT bearer token; user or admin role.                      |
| products                       | Course catalogue and vector-sync state.                  | Written first to SQL; semantic copy upserted; failed sync is recoverable. |
| events                         | Meaningful learner actions and metadata.                 | Batched async intake; short dwell retained but not always agent-eligible. |
| recommendations + items        | Narrative and ranked, real-course selections.            | New result becomes active; previous result is retained as history.        |
| agent_runs + steps + Mesh logs | Traceable decision and model-use evidence.               | Admin-only review surface; measured latency and token metadata.           |

### Privacy and production guardrails

Tracking consent is explicit at sign-up. A learner may opt out, which stops client-side event queuing and prevents recommendation triggering without preventing ordinary use of the marketplace. The collector avoids raw scroll and mousemove streams; it records meaningful actions and one computed time-on-page event instead. Admin views are role-gated, and learner emails shown in operations are confined to that admin-only interface.

### LLM-call efficiency

The recommendation endpoint first asks whether regeneration is warranted. No signal means no agent run. If an active recommendation exists, the backend requires enough new meaningful events since that recommendation and an elapsed cooldown; otherwise it returns the cached result. The explicit Refresh action is a deliberate user request, but it still requires at least some activity. This keeps the system responsive, reduces duplicate calls and preserves a stable recommendation until the learner's behaviour materially changes.

## 04 · LEARNER EXPERIENCE

## Learner journey and annotated screenshots

The learner experience is a continuous discovery loop: understand the value, create an account with consent, explore the catalog, inspect a course and return to a tailored next step. Each capture below includes a concrete reviewer action.

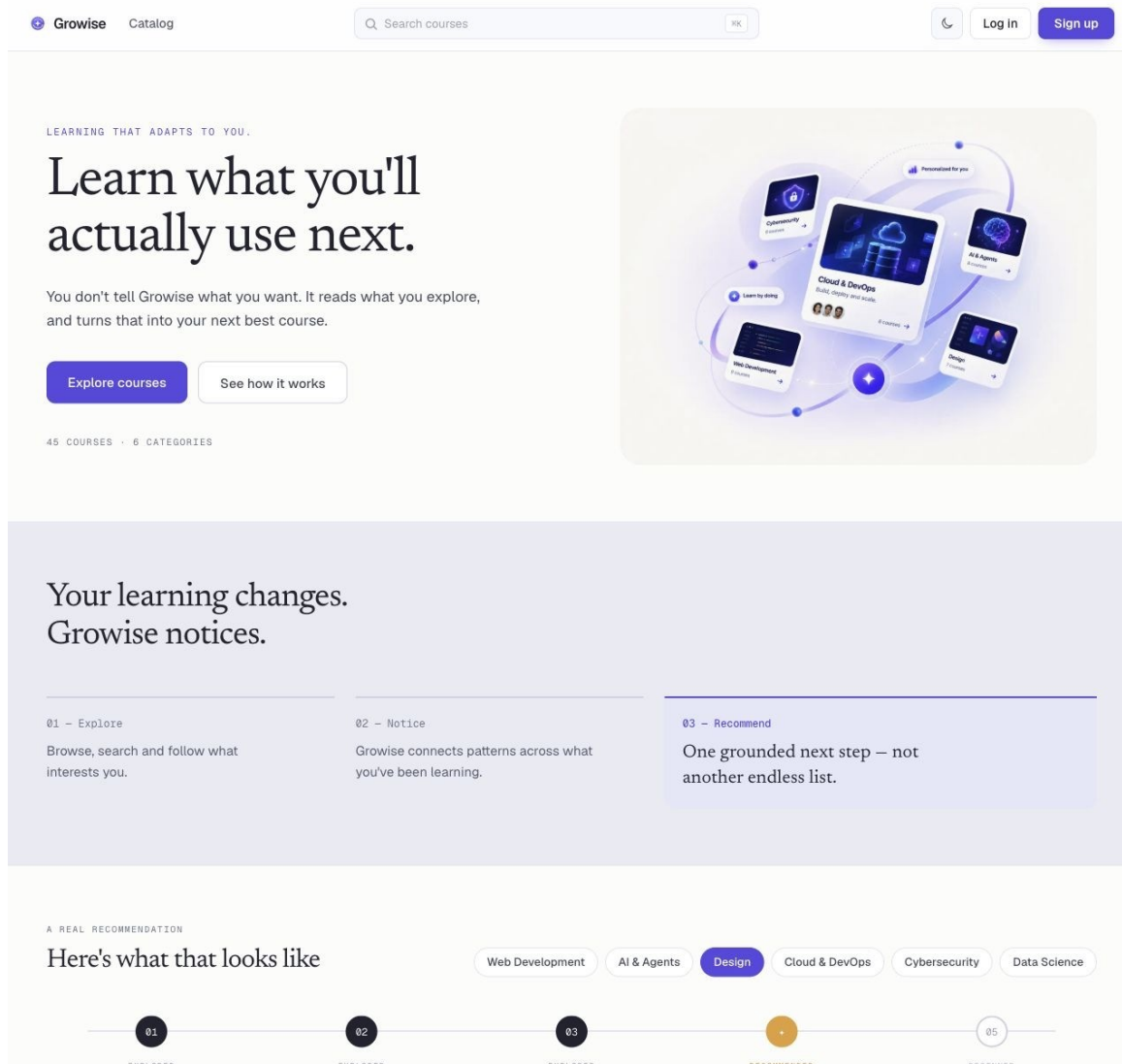


Figure 4. Learner landing page. **Reviewer action:** Select “Explore courses” to enter discovery, or use “See how it works” to connect the three-step story with the product behaviour.

Growise Catalog

Search courses

Checking session...

LEARNING THAT ADAPTS TO YOU

Growise notices what you explore and tells you what's next.

Start with the subjects that matter to you, then let your learning path become more useful with every step.

Growise

Start learning

Already have an account? [Log in](#)

Email

you@company.com

Password

☒ Let the Growise agent learn from what I browse  
Turn this off any time. Recommendations stop, nothing else changes.

Create account

Com

Figure 5. Sign-up page with tracking consent. **Reviewer action:** Create an account with the consent switch enabled to allow the recommendation agent to learn from meaningful browsing; switch it off to use Growise without recommendations.

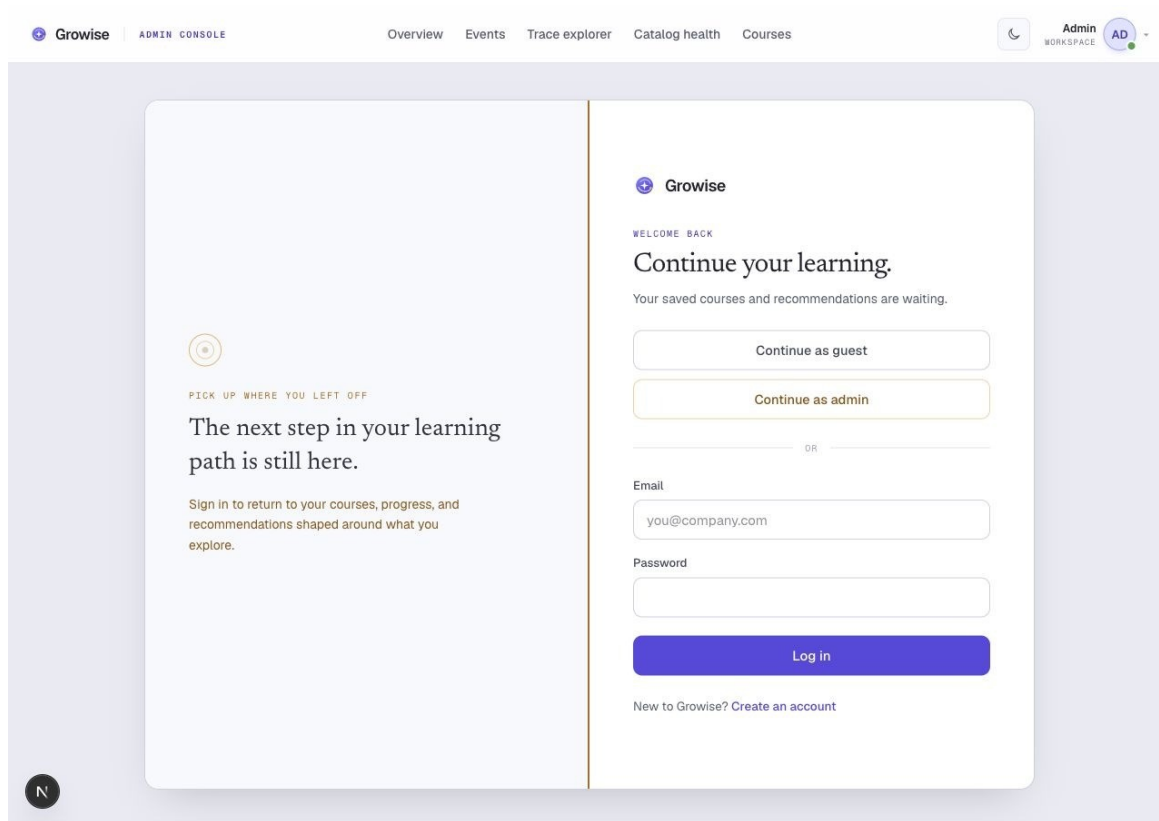


Figure 6. Learner login. **Reviewer action:** Use “Continue as guest” for the seeded learner walkthrough, or sign in with an existing email and password to restore courses and recommendations.



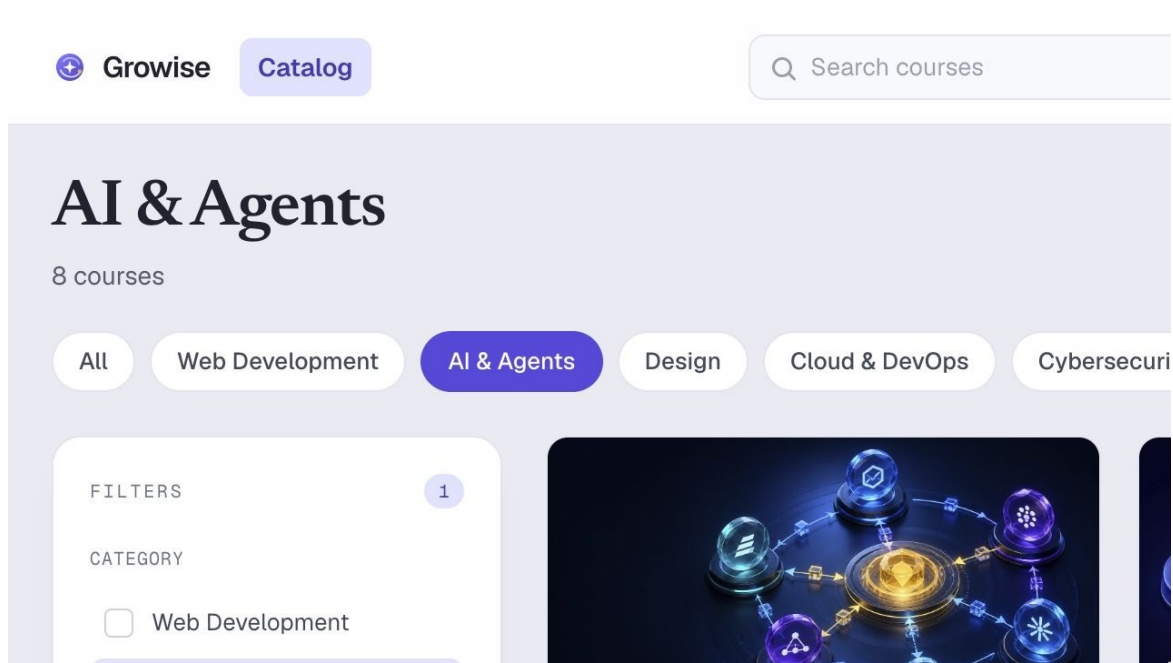


Figure 7. Filtered course catalogue. **Reviewer action:** Choose a category, level or price filter and open a course card; the resulting views and clicks become lightweight recommendation evidence after sign-in.

Growise

Catalog

Q

Search courses

✕

☾

Log in

Sign up

CATALOG / AI & AGENTS

AI & AGENTS

INTERMEDIATE

# Production LangGraph: Agents That Ship

Build, evaluate and deploy multi-step agents that survive real traffic — retries, human handoff, cost ceilings, and the observability to prove any of it works.

Dr. Amara Osei ★★★★★ 4.8 (2,431 REVIEWS) · 12H 40M · 48 LESSONS

langgraph agents production observability

INSIDE THE COURSE

## Make Production LangGraph: Agents That Ship a capability your team can rely on.

Build, evaluate and deploy multi-step agents that survive real traffic — retries, human handoff, cost ceilings, and the observability to prove any of it works. This guided path moves from first principles to applied decisions, so you finish with techniques you can use in real work.

- ✓ Build a repeatable approach to agent architecture and state.
- ✓ Make sound trade-offs across tools, memory and handoffs.
- ✓ Apply evaluation and observability in hands-on project work.
- ✓ Leave with a practical playbook for deployment and operational guardrails.

COURSE CURRICULUM

## A practical path, section by section.

4 SECTIONS · 48 LESSONS

01

### Agent architecture and state

Build practical judgment around agent architecture and state through focused e...

12 lessons · 3H 10M

AGENT WORKFLOWS  
COURSE PREVIEW

ONE-TIME ENROLLMENT

**\$79** ~~\$149~~

Enroll now

LOG IN TO ENROLL

AT A GLANCE

|                     |                       |
|---------------------|-----------------------|
| DURATION            | LESSONS               |
| <b>12H 40M</b>      | <b>48 lessons</b>     |
| LEVEL               | INSTRUCTOR            |
| <b>Intermediate</b> | <b>Dr. Amara Osei</b> |

Preview the curriculum

INCLUDED WITH ENROLLMENT

- ✓ Lifetime access
- ✓ 48 on-demand video lessons
- ✓ Certificate of completion

**Your Signal**

LIVE · OBSERVED BY AGENT

Sign in to personalise this course with your learning signal.

Learn at your own pace, with full access from the moment you enroll.

Figure 8. Course detail page. **Reviewer action:** Review the outcomes and expandable curriculum, then use “Enroll now” when ready; this is also the place where a signed-in learner sees the current learning signal.

Growise · SmartReco Hackathon · Page 10

Growise
Catalog
For you
Search courses
My learning
TA

TUESDAY 11 AUGUST

# For you, Taylor

YOUR LEARNING PATH · updated 18 min ago
Update path

To build on your strong interest in AI & Agents and your recent engagement in design, consider taking the Multi-Agent Systems in Practice course to deepen your understanding of agent coordination. Additionally, the Design for Accessibility course can enhance your design skills by integrating accessibility principles into your work.

WHAT SHAPED THIS PATH

Strongest theme: AI & Agents
You spent 58s exploring Design Systems for Product Teams
Keeping Evaluating LLM Systems out based on your feedback

► How your focus is calculated

Courses already in your learning library, and any course you dismiss, stay out of future picks.

RECOMMENDED FOR YOU

A few courses selected from what you've been exploring recently.



**MULTI-AGENT**

ADVANCED

**Multi-Agent Systems in Practice**

Dr. Amara Osei · ★ 4.9 (418)

11h 30m · 42 LESSONS



**ACCESSIBLE DESIGN**

INTERMEDIATE

**Design for Accessibility**

Jonas Weber · ★ 4.8 (512)

5h 55m · 22 LESSONS

Figure 9. Personalised “For you” page. **Reviewer action:** Review the narrative and “Why this fits” explanations, then choose a recommended course or use “Not for me” to exclude a poor fit from future recommendations.

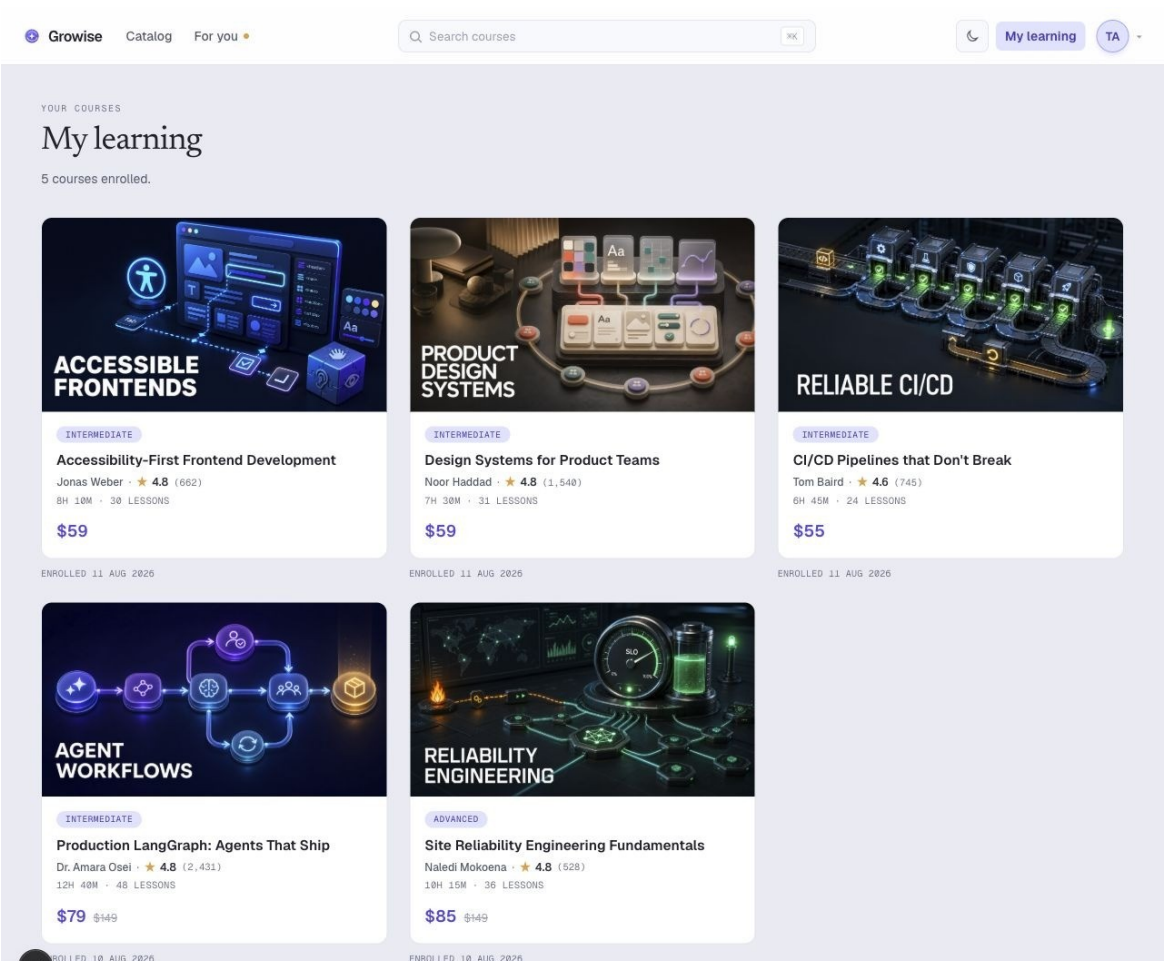


Figure 10. My learning library. **Reviewer action:** Open an enrolled course to continue learning; courses already owned are excluded from future recommendation candidates.

## 05 · ADMINISTRATOR EXPERIENCE

## Admin journey and annotated screenshots

The admin console makes the recommendation system inspectable rather than magical. It is where a reviewer can confirm the operating health of the catalog, inspect consented signals and audit the exact decision path behind a learner-facing recommendation.

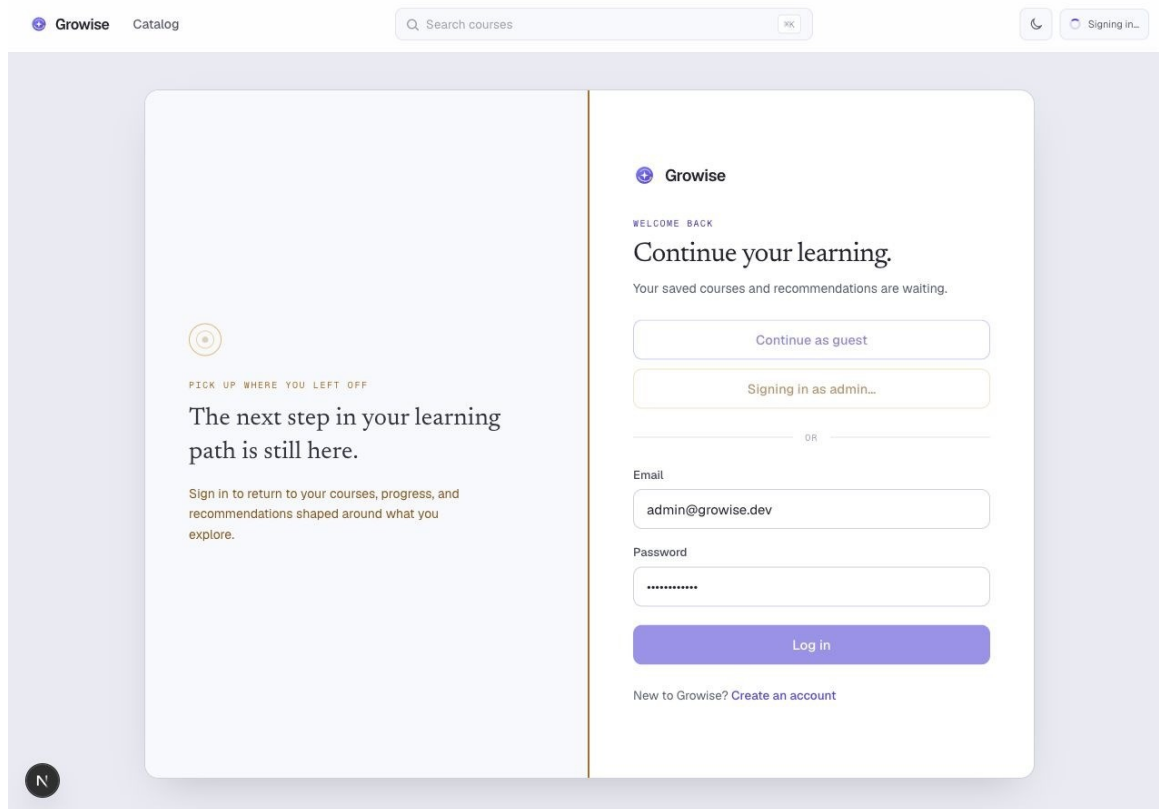


Figure 11. Administrator login path. **Reviewer action:** Choose “Continue as admin” on the sign-in screen to enter the seeded operational workspace and land on the recommendation dashboard.

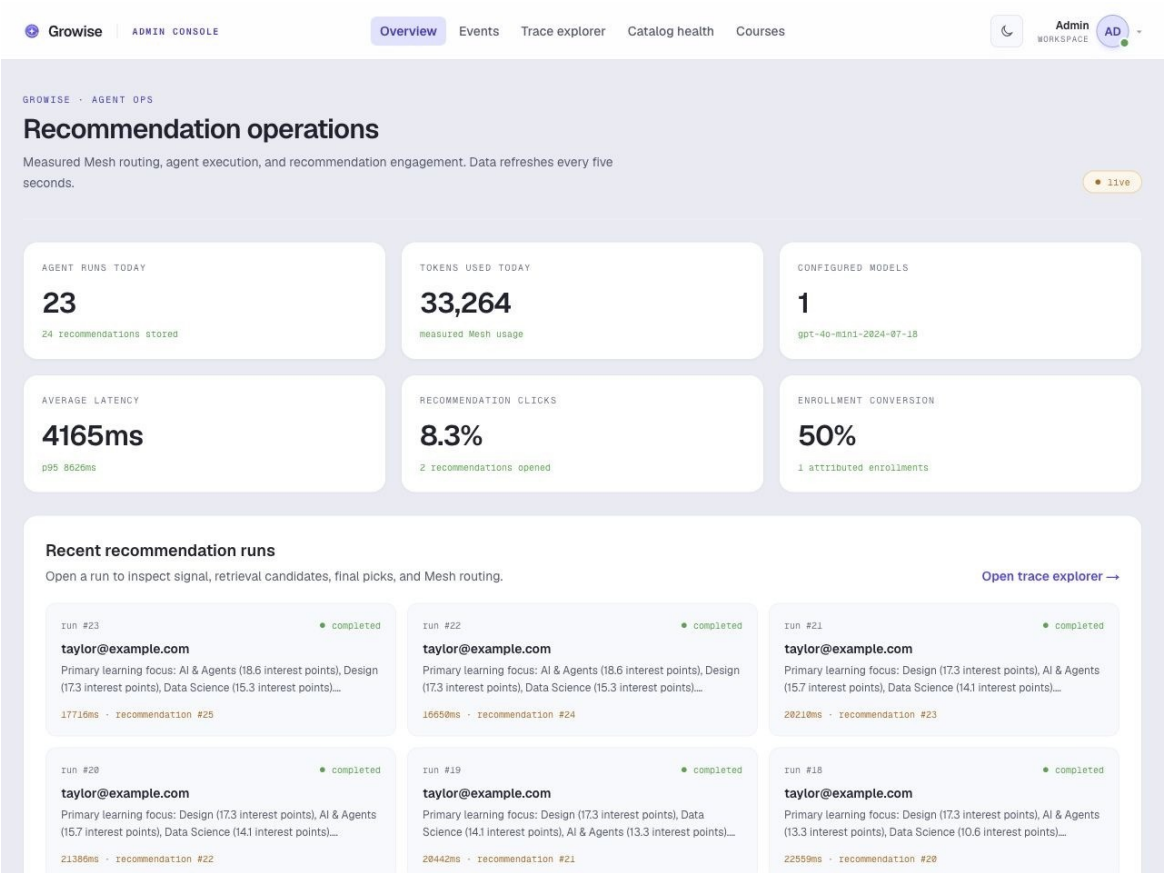


Figure 12. Recommendation operations dashboard. **Reviewer action:** Start here to review daily run volume, Mesh tokens, latency and recommendation engagement; open a recent run for its underlying trace.

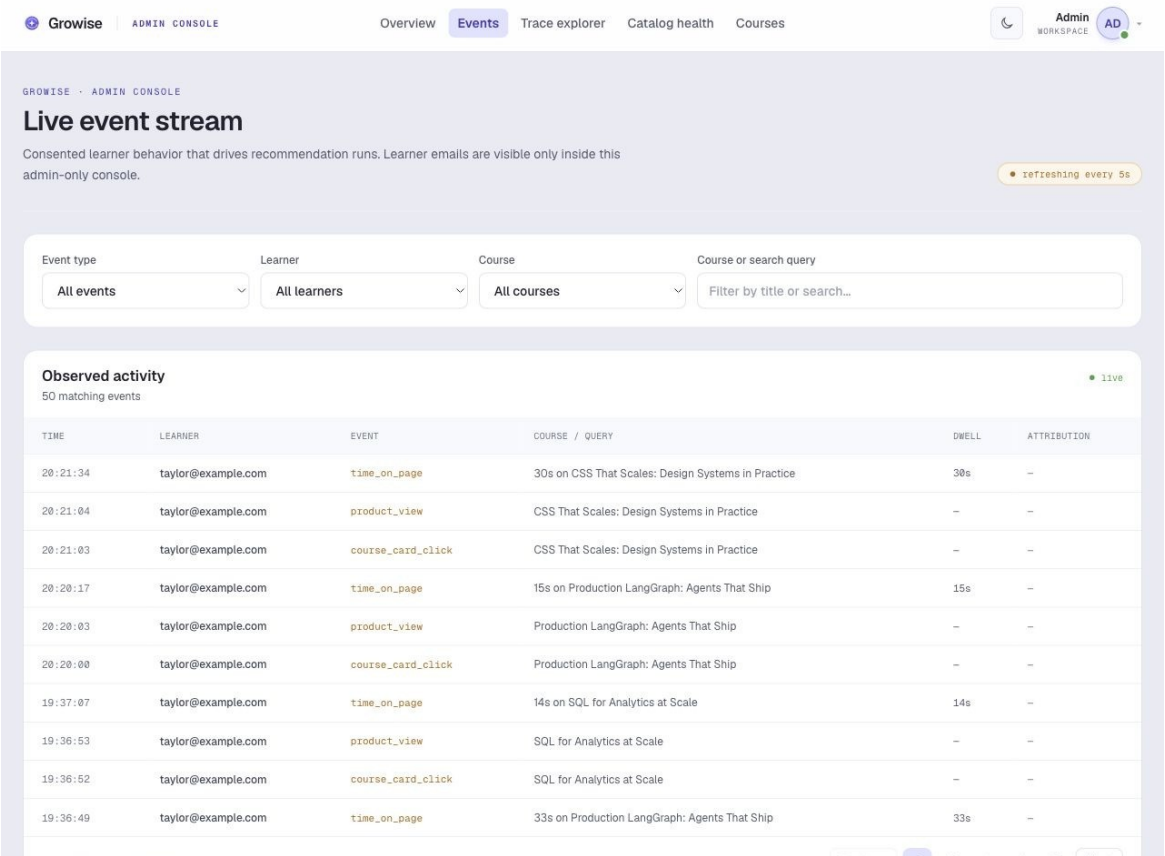


Figure 13. Live consented event stream. **Reviewer action:** Filter by event type, learner, course or search query to check the behavioural evidence available to the system.

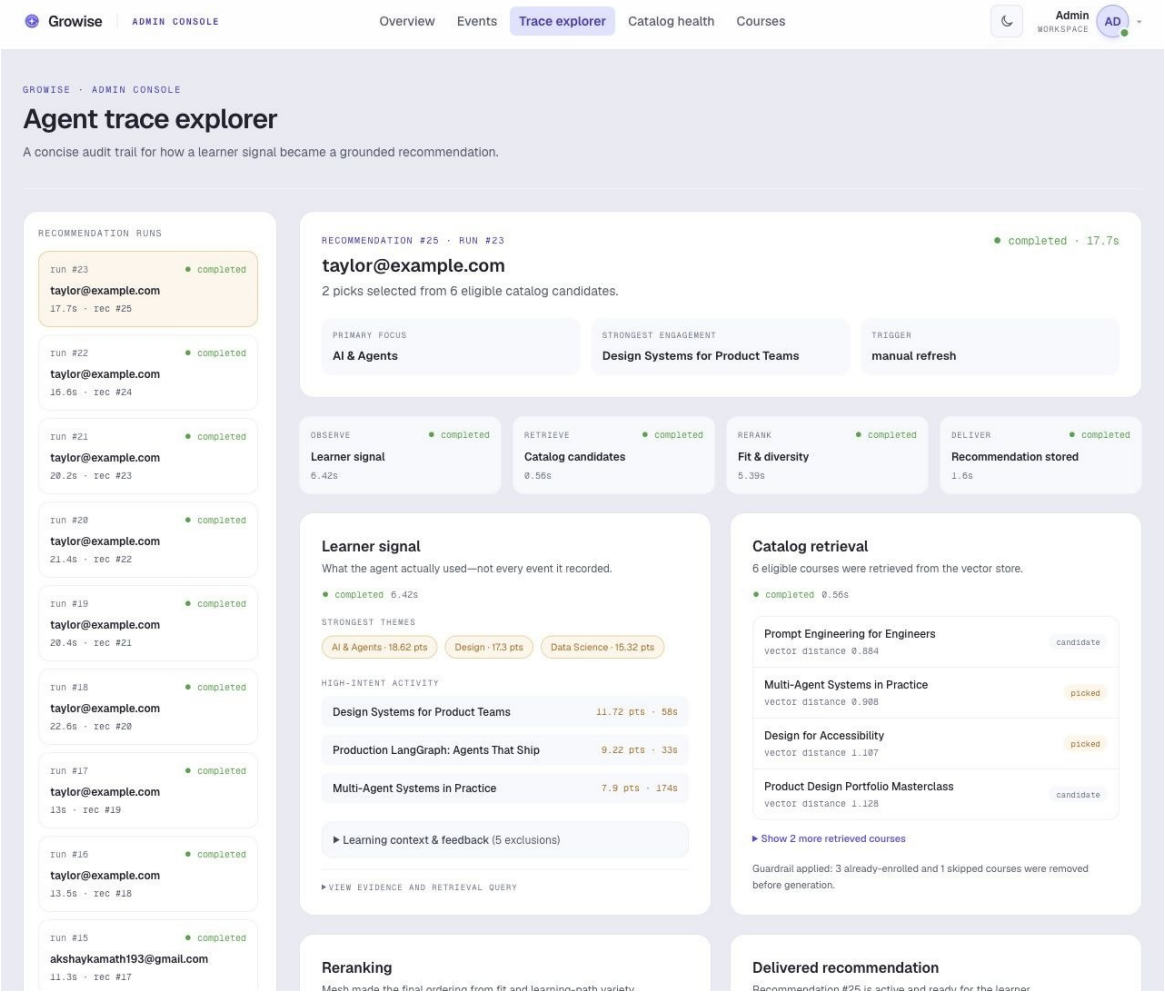


Figure 14. Agent trace explorer. **Reviewer action:** Select a recommendation run to verify the learner signal, retrieved candidates, exclusions, reranking, delivered courses and measured Mesh calls.



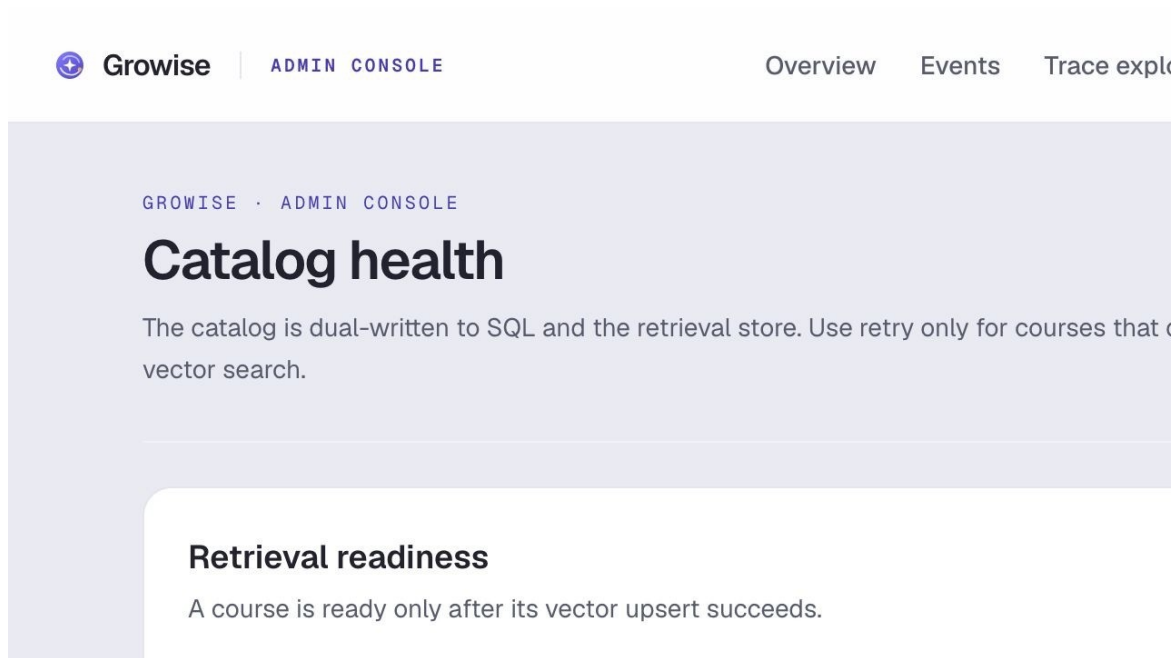


Figure 15. Catalog health. **Reviewer action:** Confirm retrieval coverage is 100%; if a course is unsynced, use “Retry unsynced courses” to repair only the affected vector records.

Growise

ADMIN

CONSOLE

Overview

Events

Trace explorer

Catalog health

Courses

Admin

WORKSPACE

AD

ADMIN

Course management

+ Add course

| TITLE                                    | CATEGORY        | LEVEL        | PRICE | SYNCED                       |                                   |
|------------------------------------------|-----------------|--------------|-------|------------------------------|-----------------------------------|
| Platform Engineering Foundations         | Cloud & DevOps  | Intermediate | \$69  | <div><div></div>synced</div> | <div>Edit</div> <div>Delete</div> |
| Accessibility-First Frontend Development | Web Development | Intermediate | \$59  | <div><div></div>synced</div> | <div>Edit</div> <div>Delete</div> |
| Web Testing with Playwright              | Web Development | Intermediate | \$55  | <div><div></div>synced</div> | <div>Edit</div> <div>Delete</div> |
| Agent Security & Guardrails              | AI & Agents     | Intermediate | \$69  | <div><div></div>synced</div> | <div>Edit</div> <div>Delete</div> |
| Cryptography for Developers              | Cybersecurity   | Intermediate | \$55  | <div><div></div>synced</div> | <div>Edit</div> <div>Delete</div> |
| Incident Response Playbooks              | Cybersecurity   | Advanced     | \$75  | <div><div></div>synced</div> | <div>Edit</div> <div>Delete</div> |
| Security for API Backends                | Cybersecurity   | Intermediate | \$59  | <div><div></div>synced</div> | <div>Edit</div> <div>Delete</div> |
| Ethical Hacking & Penetration Testing    | Cybersecurity   | Advanced     | \$89  | <div><div></div>synced</div> | <div>Edit</div> <div>Delete</div> |
| Threat Modeling in Practice              | Cybersecurity   | Intermediate | \$65  | <div><div></div>synced</div> | <div>Edit</div> <div>Delete</div> |
| Application Security Fundamentals        | Cybersecurity   | Beginner     | \$49  | <div><div></div>synced</div> | <div>Edit</div> <div>Delete</div> |
| Visual Design Fundamentals               | Design          | Beginner     | \$39  | <div><div></div>synced</div> | <div>Edit</div> <div>Delete</div> |
| Figma to Production Handoff              | Design          | Beginner     | \$35  | <div><div></div>synced</div> | <div>Edit</div> <div>Delete</div> |

Showing 1-12 of 45 courses

Previous

1

2

3

4

Next

Figure 16. Course management. **Reviewer action:** Use “Add course” or “Edit” to maintain the catalog; check the Synced column after each change to confirm semantic retrieval readiness.

## 06 · REVIEWER QUICK PATH

## Suggested end-to-end review

A concise demo sequence proves both the learner value and the operational controls. First, use the landing page to explain the promise. Next, sign in as the seeded learner, browse courses in one theme, open a few details and view the “For you” page. Finally, sign in as the administrator and use Agent Ops, Events, Trace Explorer and Catalog health to show that the recommendation is grounded, observable and operationally safe.

| Step | Reviewer action                               | Expected evidence                                                                   |
|------|-----------------------------------------------|-------------------------------------------------------------------------------------|
| 1    | Open the landing page and catalogue.          | Clear value proposition, filters and real catalog.                                  |
| 2    | Sign in as guest and explore several courses. | Consented views, clicks and dwell data appear in the live event stream.             |
| 3    | Open “For you” or choose Update path.         | A short personalised narrative and real course recommendations.                     |
| 4    | Sign in as admin and open the latest trace.   | Signal, retrieval candidates, exclusions, ranking, recommendation and Mesh metrics. |
| 5    | Open Catalog health and Course management.    | Dual-write health, safe retry control and course-level sync status.                 |

### What makes this submission complete

Growise delivers the SmartReco brief as a working platform rather than a static recommendation widget: authentication and role separation, course CRUD with durable vector sync state, efficient behavioural tracking, a structured LangGraph workflow through Mesh, catalog-grounded personalised copy, stored recommendation history and a full administrator audit surface. The screenshots in this document are captured from the live local application and map directly to reviewer actions.